

No. of Functional Features Included in the Solution

Date	2 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Soil Metric Intake Form	Interactive UI to capture N,P,pH, Temp, humidity, rainfall	Static Front-End	Done	0.5 Marks
2	Input Validation Layer	Server-side boundary checking for 7 physiological soil metrics	Flask Controller (app.py)	Done	0.5 Marks
3	Multi-Variant AI Engine	Mathematical optimization matrix to compute crop recommendations	Core AI Engine	In Progress	1.5 Marks
4	Confidence Score Logic	Algorithmic scoring evaluating prediction variance limits	Core AI Engine	In Progress	0.5 Marks
5	Data Persistence Pipeline	Historical logging of multi-variant metrics to SQL store	Database Backend	Done	0.5 Marks
6	Response Builder Engine	Dynamic serialization mapping records to JSON/HTML format	Core Backend	In Progress	0.5 Marks
7	Historical Reports Grid	Admin panel viewing past prediction telemetry runs	UI & Support Core	Pending	0.5 Marks
8	Database Migration Tool	2 (Persistence, Migration Tool)	SQL Storage, Schema Setup	Done	0.5 Marks

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	5
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	4

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2 (Soil Input Form, Reports Grid)	Input UI, Admin Reports
2	Backend / Logic	3 (Validation, AI Engine, Response Builder)	Flask Logic, ML Processing
3	Database / Storage	2 (Persistence, Migration Tool)	SQL Storage, Schema Setup
4	API / Integration	1 (Flask API)	Data Communication
5	Security / Authentication	0	Not Implemented